

A ZFC Almost Disjoint Family of Size $2^{\aleph_1}$ on $\mathbb{N}^*$

Verification of the supplied construction and an exploratory full proof

22 June 2026

Abstract

Lambie-Hanson and Schritterser ask whether ZFC proves the existence of an almost disjoint family on $\mathbb{N}^* = \beta\mathbb{N} \setminus \mathbb{N}$ of cardinality $2^{\aleph_1}$, or even of the maximal possible cardinality $2^{2^{\aleph_0}}$. First, the continuum-sized construction supplied with the question is verified; in particular, its regular-open argument is correct. Second, a Boolean-algebraic construction is given which appears to answer the first part of the question positively in ZFC: there is an almost disjoint family on $\mathbb{N}^*$ of cardinality $2^{\aleph_1}$. The proof uses the completion of $\mathcal{P}(\omega)/\text{Fin}$, two elementary countable fusion/interpolation facts, and a binary interval tree of height ω_1 . The maximal target $2^{2^{\aleph_0}}$ is obtained whenever $2^{\aleph_1} = 2^{2^{\aleph_0}}$, but is not settled here in the remaining cardinal-arithmetic cases. The new proof has not been peer reviewed and should be checked independently before being cited as a definitive resolution.

Packet status

Run `fa_banach_001`.

Source arXiv id
2606.02040.

Source paper C. Lambie-Hanson and D. Schritterser, *Generalized almost disjoint families and injective Banach spaces*, arXiv:2606.02040v1.

Packet classification

Candidate full solution for the first cardinal target of Question 4.3, likely valid subject to human mathematical review.

Upgrade note This packet supersedes the earlier ZFC continuum-sized partial packet and absorbs the earlier branch-regular-tree conditional reduction. It proves the $2^{\aleph_1}$ target in ZFC. The stronger maximal target $2^{2^{\aleph_0}}$ remains open in the case $2^{\aleph_1} < 2^{2^{\aleph_0}}$.

The source question appears in Question 4.3 of Lambie-Hanson–Schritterser:

operator $r : c_0(\omega_1) \rightarrow \ell_\infty([\omega]^-)$: in [4, Theorem 1], it is proven that $\mathfrak{c} \nVdash$ implies that $\text{Ext}^2(c_0(\omega_1), c_0) \neq 0$. It is at least conceivable that the opposite answer might follow from an assumption such as the Proper Forcing Axiom.

Finally, we wonder whether an assumption like $\mathfrak{b} = 2^{\aleph_0}$ is necessary to construct a large almost disjoint family on $\mathbb{N}^*$.

Question 4.3. *Is it provable in ZFC that there exists an almost disjoint family on $\mathbb{N}^*$ of cardinality $2^{\aleph_1}$, or even of cardinality $2^{2^{\aleph_0}}$?*

Status of the literature search. The source preprint was submitted on 1 June 2026 and, in its current first version, states the question below as open. Searches through 22 June 2026 for the exact question, title, arXiv identifier, author pages, and closely related Boolean-algebraic and topological phrases did not locate a later proof or counterexample. Because the preprint is very recent, this is evidence rather than a guarantee that no independent solution exists.

1 Proof Intuition

The key step is to replace the earlier conditional branch-regular-tree hypothesis by an explicit interval tree in $\mathcal{P}(\omega)/\text{Fin}$. The countable lower-bound and interpolation lemmas provide exactly the amount of fusion needed at each countable limit level below ω_1 . This produces nested Boolean intervals whose sibling upper endpoints meet only in the parent lower endpoint.

After passing to the Boolean completion, equivalently the regular-open algebra of $\mathbb{N}^*$, each branch $f \in 2^{\omega_1}$ yields the join

$$u_f = \bigvee_{\alpha < \omega_1} \ell_{f \upharpoonright \alpha}.$$

Two different branches meet at the lower endpoint of their last common node, which is an element of $\mathcal{P}(\omega)/\text{Fin}$, hence a compact clopen intersection in $\mathbb{N}^*$. Except for at most $\mathfrak{c}$ branch joins already lying in $\mathcal{P}(\omega)/\text{Fin}$, these joins are nonclopen regular open sets and therefore noncompact. If $2^{\aleph_1} = \mathfrak{c}$, the earlier continuum construction already reaches the same cardinal.

2 The source question

For a topological space X , Lambie-Hanson and Schrittemser call a family $\mathcal{A}$ an *almost disjoint family on X* if every member of $\mathcal{A}$ is a noncompact regular open subset of X , and the intersection of every two distinct members is compact [1, Definition 2.1]. Their Question 4.3 is:

Is it provable in ZFC that there exists an almost disjoint family on $\mathbb{N}^*$ of cardinality $2^{\aleph_1}$, or even of cardinality $2^{2^{\aleph_0}}$?

They prove the first cardinality from the additional hypothesis $\mathfrak{b} = \mathfrak{c}$ [1, Theorem C].

3 Verification of the supplied continuum construction

For an infinite $a \subseteq \omega$, put

$$N_a = \{p \in \mathbb{N}^* : a \in p\}.$$

This is the basic clopen subset of $\mathbb{N}^*$ corresponding to the class of a modulo finite sets.

Proposition 3.1. *In ZFC there is an almost disjoint family on $\mathbb{N}^*$ of cardinality $\mathfrak{c} = 2^{\aleph_0}$. Its members can in fact be chosen pairwise disjoint.*

Proof. Identify $2^{<\omega}$ with ω . For $x \in 2^\omega$, let

$$A_x = \{x \upharpoonright n : n < \omega\}.$$

The family $\{A_x : x \in 2^\omega\}$ is a classical almost disjoint family on ω of cardinality $\mathfrak{c}$. Partition each A_x as

$$A_x = \bigsqcup_{n < \omega} A_{x,n},$$

where every $A_{x,n}$ is infinite, and for $n \geq 1$ set

$$b_{x,n} = \bigcup_{m < n} A_{x,m}, \quad U_x = \bigcup_{n \geq 1} N_{b_{x,n}}.$$

Clearly U_x is open and nonempty.

We verify regularity. Fix $p \notin U_x$ and a basic neighborhood $N_y \ni p$, where $y \in p$. Since $b_{x,n} \notin p$, the set $y \setminus b_{x,n}$ is infinite for every n ; otherwise $y \subseteq^* b_{x,n}$, and the nonprincipal ultrafilter p would contain $b_{x,n}$.

If $y \setminus A_x$ is infinite, choose an infinite $z \subseteq y \setminus A_x$. Then $N_z \subseteq N_y$ and $N_z \cap U_x = \emptyset$. Otherwise $y \setminus A_x$ is finite. The preceding paragraph then implies that y meets infinitely many blocks $A_{x,k}$. Choose an infinite $z \subseteq y \cap A_x$ containing at most one point from each block. For every n , the intersection $z \cap b_{x,n}$ is finite, so again $N_z \subseteq N_y$ and $N_z \cap U_x = \emptyset$.

Thus every basic neighborhood of every $p \notin U_x$ contains a nonempty open subset disjoint from U_x . In particular, no such neighborhood can be contained in $\text{cl}(U_x)$, because U_x is dense in its closure. Consequently

$$\text{int cl}(U_x) \subseteq U_x.$$

The reverse inclusion holds for every open set, hence $U_x = \text{int cl}(U_x)$: U_x is regular open.

The increasing family $\{N_{b_{x,n}} : n \geq 1\}$ is an open cover of U_x with no finite subcover. Indeed, $N_{b_{x,n+1}} \not\subseteq N_{b_{x,n}}$, since a nonprincipal ultrafilter concentrated on the infinite set $A_{x,n} = b_{x,n+1} \setminus b_{x,n}$ belongs to the former clopen set and not the latter. Hence U_x is not compact.

Finally, if $x \neq y$, then $A_x \cap A_y$ is finite. Therefore $b_{x,n} \cap b_{y,m}$ is finite for all m, n , and $N_{b_{x,n}} \cap N_{b_{y,m}} = \emptyset$ in $\mathbb{N}^*$. It follows that $U_x \cap U_y = \emptyset$. The sets U_x are nonempty, so they are distinct, and the asserted family has cardinality $\mathfrak{c}$. $\square$

Remark 3.2. The only delicate point in the supplied proof is that the smaller clopen set $N_z \subseteq N_y$ need not contain the original ultrafilter p . This is not a gap: its existence shows that N_y is not contained in $\text{cl}(U_x)$, which is exactly what is needed to exclude p from $\text{int cl}(U_x)$.

4 The Boolean-algebraic translation

Let

$$\mathbb{B} = \mathcal{P}(\omega)/\text{Fin}.$$

The Stone space of $\mathbb{B}$ is $\mathbb{N}^*$, and $\mathbb{B} \cong \text{Clop}(\mathbb{N}^*)$. Let $\widehat{\mathbb{B}}$ be the Boolean completion of $\mathbb{B}$. Since $\mathbb{N}^*$ is zero-dimensional,

$$\widehat{\mathbb{B}} \cong \text{RO}(\mathbb{N}^*),$$

with $\mathbb{B}$ embedded as the clopen regular open sets. We use the same symbol for a Boolean element and the corresponding regular open set when no confusion can arise.

Lemma 4.1 (Translation lemma). *Suppose $\mathcal{U} \subseteq \widehat{\mathbb{B}} \setminus \mathbb{B}$ and $u \wedge v \in \mathbb{B}$ whenever $u, v \in \mathcal{U}$ are distinct. Then the corresponding regular open subsets of $\mathbb{N}^*$ form an almost disjoint family on $\mathbb{N}^*$.*

Proof. In the regular-open algebra, finite meets are set-theoretic intersections. Thus $u \wedge v \in \mathbb{B}$ says that the intersection is clopen, hence compact. A regular open subset of a compact Hausdorff space is compact if and only if it is clopen: compact subsets are closed, and a clopen subset is closed in a compact space. Therefore every element of $\widehat{\mathbb{B}} \setminus \mathbb{B}$ corresponds to a noncompact regular open set. $\square$

The rest of the proof is therefore purely Boolean-algebraic.

5 Two countable fusion facts in $\mathcal{P}(\omega)/\text{Fin}$

We write $[A]$ for the class modulo finite sets of $A \subseteq \omega$.

Lemma 5.1 (Countable lower bounds). *If*

$$d_0 \geq d_1 \geq d_2 \geq \cdots > 0$$

in $\mathbb{B}$, then there is $q > 0$ such that $q \leq d_n$ for every n .

Proof. Choose representatives $D_n \subseteq \omega$. Replacing D_n by $\bigcap_{i \leq n} D_i$, we may assume that the representatives are actually decreasing and infinite. Recursively choose distinct $k_n \in D_n$. Then $Q = \{k_n : n < \omega\}$ is infinite and $Q \subseteq^* D_n$ for every n , so $q = [Q]$ works. $\square$

Lemma 5.2 (Countable interpolation). *Let $x_n, y_n \in \mathbb{B}$ for $n < \omega$, and suppose*

$$x_n \leq y_m \quad \text{for all } m, n < \omega.$$

Then there is $z \in \mathbb{B}$ such that

$$x_n \leq z \leq y_m \quad \text{for all } m, n < \omega.$$

Proof. Choose representatives $X_n, Y_n \subseteq \omega$, and put

$$X'_n = \bigcup_{i \leq n} X_i, \quad Y'_n = \bigcap_{j \leq n} Y_j.$$

Then $X'_n \subseteq^* Y'_n$. Choose strictly increasing integers k_n tending to infinity such that

$$X'_n \setminus Y'_n \subseteq k_n$$

for every n , where an integer is identified with its initial segment of ω . Define

$$Z = \bigcup_{n < \omega} (X'_n \cap [k_n, k_{n+1})).$$

For fixed i , every point of X_i above k_i lies in a block indexed by some $n \geq i$, and hence belongs to Z . Thus $X_i \subseteq^* Z$. For fixed j , every block with index $n \geq j$ is contained in $Y'_n \subseteq Y_j$; the earlier blocks form a finite set. Thus $Z \subseteq^* Y_j$. Therefore $z = [Z]$ interpolates the two countable families. $\square$

6 A binary interval tree of height ω_1

For two binary sequences s, t , write $s \wedge t$ for their longest common initial segment.

Lemma 6.1 (Interval tree). *There are elements*

$$\langle \ell_s, r_s : s \in 2^{<\omega_1} \rangle \subseteq \mathbb{B}$$

with the following properties.

(i) $\ell_s < r_s$ for every s .

(ii) If $s \subseteq t$, then

$$\ell_s \leq \ell_t < r_t \leq r_s.$$

(iii) For every s and $i < 2$, $\ell_s < \ell_{s \smallfrown \langle i \rangle}$.

(iv) If s, t are incomparable, then

$$r_s \wedge r_t = \ell_{s \wedge t}.$$

Proof. We construct the intervals $[\ell_s, r_s]$ recursively by levels. Start with $\ell_\emptyset = 0$ and $r_\emptyset = 1$.

Suppose $\ell_s < r_s$ has been defined, and let

$$d = r_s \wedge \neg \ell_s > 0.$$

The algebra $\mathbb{B}$ is atomless, so split d into four pairwise disjoint nonzero elements p_0, q_0, p_1, q_1 whose join is d . For $i < 2$, set

$$\ell_{s \smallfrown \langle i \rangle} = \ell_s \vee p_i, \quad r_{s \smallfrown \langle i \rangle} = \ell_s \vee p_i \vee q_i.$$

Then each child interval is nondegenerate and lies inside the parent interval, its lower endpoint is strictly above ℓ_s , and

$$r_{s \smallfrown \langle 0 \rangle} \wedge r_{s \smallfrown \langle 1 \rangle} = \ell_s.$$

Now let $t \in 2^\lambda$, where $\lambda < \omega_1$ is a nonzero limit ordinal, and suppose all proper initial-segment intervals have been defined. Choose a strictly increasing cofinal sequence $\langle \alpha_n : n < \omega \rangle$ in λ , and put $t_n = t \upharpoonright \alpha_n$. The elements

$$d_n = r_{t_n} \wedge \neg \ell_{t_n}$$

form a decreasing sequence of nonzero elements. By Lemma 5.1, choose $q > 0$ with $q \leq d_n$ for every n . Put

$$e_n = r_{t_n} \wedge \neg q.$$

For all m, n , nesting of the earlier intervals gives $\ell_{t_m} \leq r_{t_n}$, while $q \leq \neg \ell_{t_m}$. Hence $\ell_{t_m} \leq e_n$. By Lemma 5.2, choose $z \in \mathbb{B}$ such that

$$\ell_{t_m} \leq z \leq e_n \quad (m, n < \omega).$$

Define

$$\ell_t = z, \quad r_t = z \vee q.$$

Since $z \wedge q = 0$ and $q > 0$, this interval is nondegenerate. Moreover, for every n ,

$$\ell_{t_n} \leq \ell_t < r_t \leq (r_{t_n} \wedge \neg q) \vee q = r_{t_n}.$$

Cofinality of the sequence then gives nesting inside every proper initial-segment interval.

This completes the recursion. Finally, if s, t are incomparable and $u = s \wedge t$, then s and t extend different immediate successors of u . By nesting, their upper endpoints lie below the two corresponding sibling upper endpoints, whose meet is ℓ_u . Conversely $\ell_u \leq \ell_s, \ell_t \leq r_s, r_t$. Hence $r_s \wedge r_t = \ell_u$, proving (iv). $\square$

7 The ZFC construction of size $2^{\aleph_1}$

Let $\widehat{\mathbb{B}}$ be the completion of $\mathbb{B}$. For each branch $f \in 2^{\omega_1}$, define

$$u_f = \bigvee_{\alpha < \omega_1} \ell_{f \upharpoonright \alpha} \quad \text{in } \widehat{\mathbb{B}}.$$

Lemma 7.1. *If $f, g \in 2^{\omega_1}$ are distinct, δ is the least ordinal at which they differ, and $s = f \restriction \delta = g \restriction \delta$, then*

$$u_f \wedge u_g = \ell_s \in \mathbb{B}.$$

In particular, the elements u_f are pairwise distinct.

Proof. In a complete Boolean algebra, meet with a fixed element preserves arbitrary joins. Therefore

$$u_f \wedge u_g = \bigvee_{\alpha, \beta < \omega_1} (\ell_{f \restriction \alpha} \wedge \ell_{g \restriction \beta}).$$

Every term in this join is at most ℓ_s . Indeed, if at least one of α, β is at most δ , this follows from nesting along the common branch. If both are greater than δ , the two nodes are incomparable with common initial segment s ; by Lemma 6.1(iv),

$$\ell_{f \restriction \alpha} \wedge \ell_{g \restriction \beta} \leq r_{f \restriction \alpha} \wedge r_{g \restriction \beta} = \ell_s,$$

while ℓ_s lies below both lower endpoints, so equality holds. Hence the displayed supremum is exactly ℓ_s .

By Lemma 6.1(iii), $u_f \geq \ell_{f \restriction (\delta+1)} > \ell_s$, and similarly $u_g > \ell_s$. Thus $u_f = u_g$ would contradict $u_f \wedge u_g = \ell_s$, proving distinctness. $\square$

Theorem 7.2. *ZFC proves that there is an almost disjoint family on $\mathbb{N}^*$ of cardinality $2^{\aleph_1}$.*

Proof. Let $\lambda = 2^{\aleph_1}$. Since $|2^{<\omega_1}| = \mathfrak{c}$, Lemma 6.1 is a construction of only $\mathfrak{c}$ many elements of $\mathbb{B}$, as expected. By Lemma 7.1, the map

$$f \mapsto u_f \quad (f \in 2^{\omega_1})$$

is injective and the meet of any two distinct values belongs to $\mathbb{B}$.

Recall that $|\mathbb{B}| = \mathfrak{c}$ and that $\mathfrak{c} \leq 2^{\aleph_1} = \lambda$. If $\lambda > \mathfrak{c}$, then at most $\mathfrak{c}$ of the λ distinct elements u_f can lie in $\mathbb{B}$. After discarding those, there remain λ elements of $\widehat{\mathbb{B}} \setminus \mathbb{B}$ with pairwise meets in $\mathbb{B}$. The translation lemma produces the required almost disjoint family.

If instead $\lambda = \mathfrak{c}$, Proposition 3.1 already provides an almost disjoint family of cardinality $\mathfrak{c} = 2^{\aleph_1}$. These two cases exhaust ZFC. $\square$

8 Consequences and the remaining maximal target

The regular-open algebra has cardinality exactly $2^{\mathfrak{c}}$. The upper bound follows because $\mathbb{N}^*$ has a clopen base of size $\mathfrak{c}$. For the lower bound, $\mathbb{B}$ has an antichain of size $\mathfrak{c}$, obtained from a classical continuum-sized almost disjoint family on ω , and distinct subfamilies of that antichain have distinct joins in $\widehat{\mathbb{B}}$. Thus $2^{\mathfrak{c}}$ is indeed the maximal possible size in Question 4.3.

Corollary 8.1. *If*

$$2^{\aleph_1} = 2^{\mathfrak{c}},$$

then there is an almost disjoint family on $\mathbb{N}^$ of maximal size $2^{\mathfrak{c}} = 2^{2^{\aleph_0}}$.*

Proof. Apply Theorem 7.2. $\square$

Combining Theorem 7.2 with Theorem B of Lambie-Hanson–Schrutens also gives the following conditional Banach-space consequence.

Corollary 8.2. *If*

$$2^{2^{\aleph_1}} > 2^{\mathfrak{c}},$$

then the quotient $\ell_\infty([\omega]^\omega)/C(\mathbb{N}^)$ is not injective, and hence $\text{id}(c_0) \geq 3$.*

Proof. Use Theorem 7.2 with $\lambda = 2^{\aleph_1}$, the dense set of $\mathbb{N}^*$ of cardinality $\mathfrak{c}$ used in [1], and [1, Theorem B]. $\square$

The argument above does not settle the maximal target in models satisfying

$$2^{\aleph_1} < 2^{\mathfrak{c}}.$$

Its natural stopping point is ω_1 : every ordinal below ω_1 has countable cofinality, so the limit step is handled by the two countable fusion lemmas. A direct extension to a tree of height $\mathfrak{c}$ would require interpolation across limits of uncountable cofinality, which is not provided by the elementary structure of $\mathcal{P}(\omega)/\text{Fin}$.

There is also a simple obstruction to one tempting shortcut. Starting with an antichain $\{a_\xi : \xi < \mathfrak{c}\} \subseteq \mathbb{B}$, one might take joins indexed by an ordinary family of subsets of $\mathfrak{c}$ having pairwise finite intersections. Such a family cannot have size greater than $\mathfrak{c}$: choose a countably infinite subset of each member; these choices are distinct, while $|[\mathfrak{c}]^\omega| = \mathfrak{c}^\omega = \mathfrak{c}$. A maximal construction therefore needs more structure than the naive finite-intersection coding. This observation does not rule out a different ZFC proof of the maximal target.

9 Audit summary

1. The supplied continuum-sized construction is correct. The regular-open verification works exactly because every neighborhood outside U_x contains a nonempty open subset disjoint from U_x .
2. No literature resolution later than the 1 June 2026 source preprint was located in the search described above.
3. Theorem 7.2 gives a candidate full ZFC answer to the first cardinal target $2^{\aleph_1}$. Its nonstandard ingredient is the interval-tree lemma; all fusion and interpolation steps are proved explicitly.
4. The maximal target $2^{2^{\aleph_0}}$ remains open in the cardinal arithmetic case $2^{\aleph_1} < 2^{2^{\aleph_0}}$, at least by the methods developed here.

References

- [1] C. Lambie-Hanson and D. Schritterser, *Generalized almost disjoint families and injective Banach spaces*, arXiv:2606.02040v1, 2026. <https://arxiv.org/abs/2606.02040>
- [2] R. Sikorski, *Boolean Algebras*, third edition, Springer, 1969.
- [3] S. Koppelberg, General theory of Boolean algebras, in *Handbook of Boolean Algebras*, Vol. 1, North-Holland, 1989, pp. 1–307.